

10.7.90

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Question

A circle touches the line $y = x$ at a point P such that $OP = 4\sqrt{2}$, where O is the origin. The circle contains the point $(-10, 2)$ in its interior and the length of its chord on the line $x + y = 0$ is $6\sqrt{2}$. Determine the equation of the circle.

Theoretical solution

Let the point of tangency be $\mathbf{P}$. Since it lies on the line $y = x$, its coordinates must be equal.

$$\mathbf{P} = \begin{pmatrix} p \\ p \end{pmatrix} \quad (0.1)$$

The distance from the origin is given as $\|\mathbf{P}\| = 4\sqrt{2}$.

$$\begin{aligned} \|\mathbf{P}\| &= \sqrt{p^2 + p^2} = |p|\sqrt{2} = 4\sqrt{2} \\ \implies |p| &= 4 \end{aligned} \quad (0.2)$$

This gives two possible points of tangency.

$$\mathbf{P}_1 = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{P}_2 = \begin{pmatrix} -4 \\ -4 \end{pmatrix} \quad (0.3)$$

Theoretical solution

Let the center of the circle be $\mathbf{C}$.

$$\mathbf{C} = \begin{pmatrix} h \\ k \end{pmatrix} \quad (0.4)$$

The center lies on a line perpendicular to the tangent line, which has the matrix equation $(1 \quad -1) \mathbf{x} = 0$. The normal to the tangent is $\mathbf{n}_t = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

A normal to the line of centers, $\mathbf{n}_c$, must be orthogonal to $\mathbf{n}_t$.

$$\mathbf{n}_c = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.5)$$

The equation for the line of centers is $\mathbf{n}_c^\top \mathbf{x} = d$, where $d = \mathbf{n}_c^\top \mathbf{P}$. For $\mathbf{P}_1$, the line of centers is:

$$(1 \quad 1) \mathbf{C} = (1 \quad 1) \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 8 \quad (0.6)$$

For $\mathbf{P}_2$, the line of centers is:

$$(1 \quad 1) \mathbf{C} = (1 \quad 1) \begin{pmatrix} -4 \\ -4 \end{pmatrix} = -8 \quad (0.7)$$

Theoretical solution

The circle has a chord on the line with equation $(1 \ 1) \mathbf{x} = 0$. The distance from the center $\mathbf{C}$ to this line is d_{chord} .

$$d_{chord} = \frac{|\mathbf{n}_c^\top \mathbf{C}|}{\|\mathbf{n}_c\|} = \frac{|h + k|}{\sqrt{2}} \quad (0.8)$$

The radius r , distance d_{chord} , and half the chord length ($L = 3\sqrt{2}$) are related by $r^2 = d_{chord}^2 + L^2$.

$$r^2 = \left(\frac{|h + k|}{\sqrt{2}} \right)^2 + (3\sqrt{2})^2 = \frac{(h + k)^2}{2} + 18 \quad (0.9)$$

The radius is also the distance from the center to the point of tangency, $r^2 = \|\mathbf{C} - \mathbf{P}\|^2$.

Theoretical solution

For the first case $((1 \ 1) \mathbf{C} = 8$ and $\mathbf{P} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$):

$$r^2 = \frac{8^2}{2} + 18 = 50 \quad (0.10)$$

$$\begin{aligned} \|\mathbf{C} - \mathbf{P}\|^2 &= (h - 4)^2 + (k - 4)^2 = (h - 4)^2 + (8 - h - 4)^2 = 2(h - 4)^2 = 50 \\ (h - 4)^2 &= 25 \implies h - 4 = \pm 5 \implies h = 9 \text{ or } h = -1 \end{aligned} \quad (0.11)$$

This gives centers $\mathbf{C}_A = \begin{pmatrix} 9 \\ -1 \end{pmatrix}$ and $\mathbf{C}_B = \begin{pmatrix} -1 \\ 9 \end{pmatrix}$.

For the second case $((1 \ 1) \mathbf{C} = -8$ and $\mathbf{P} = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$):

$$r^2 = \frac{(-8)^2}{2} + 18 = 50 \quad (0.12)$$

$$\begin{aligned} \|\mathbf{C} - \mathbf{P}\|^2 &= (h + 4)^2 + (k + 4)^2 = (h + 4)^2 + (-8 - h + 4)^2 = 2(h + 4)^2 = 50 \\ (h + 4)^2 &= 25 \implies h + 4 = \pm 5 \implies h = 1 \text{ or } h = -9 \end{aligned} \quad (0.13)$$

$$(h + 4)^2 = 25 \implies h + 4 = \pm 5 \implies h = 1 \text{ or } h = -9 \quad (0.14)$$

Theoretical solution

This gives centers $\mathbf{C}_C = \begin{pmatrix} 1 \\ -9 \end{pmatrix}$ and $\mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

The point $\mathbf{Q} = \begin{pmatrix} -10 \\ 2 \end{pmatrix}$ is in the circle's interior, so $\|\mathbf{Q} - \mathbf{C}\|^2 < r^2 = 50$.

$$\text{For } \mathbf{C}_A = \begin{pmatrix} 9 \\ -1 \end{pmatrix} : \left\| \begin{pmatrix} -19 \\ 3 \end{pmatrix} \right\|^2 = 370 > 50 \quad (0.15)$$

$$\text{For } \mathbf{C}_B = \begin{pmatrix} -1 \\ 9 \end{pmatrix} : \left\| \begin{pmatrix} -9 \\ -7 \end{pmatrix} \right\|^2 = 130 > 50 \quad (0.16)$$

$$\text{For } \mathbf{C}_C = \begin{pmatrix} 1 \\ -9 \end{pmatrix} : \left\| \begin{pmatrix} -11 \\ 11 \end{pmatrix} \right\|^2 = 242 > 50 \quad (0.17)$$

$$\text{For } \mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix} : \left\| \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\|^2 = 2 < 50 \quad (0.18)$$

The correct center is $\mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

The center is $\begin{pmatrix} -9 \\ 1 \end{pmatrix}$ and radius squared is $r^2 = 50$. The equation is

Theoretical solution

$$x^2 + y^2 + 18x - 2y + 32 = 0 \quad (0.19)$$

In the matrix form $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} 9 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 32 = 0 \quad (0.20)$$

Plot

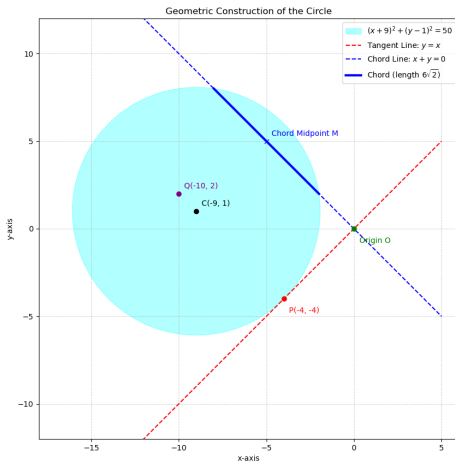


Fig :